

How to modify your Bell Touch Tone Fone
to have 1633 cycle tone
by The Magician

In issue 62 of TAP, I gave you information on the use of 1633 cycle tones for setting up toll free loop-arounds, mass conferencing, and becoming the information operator as well as applications with Autovon.

Now I will describe how to modify a standard Bell Touch Tone fone to have the extra four 1633 buttons. An excellent article on this subject appeared in the April, 1975 issue of the now defunct "TEL" magazine. The modification is very simple and requires only a SPDT toggle switch (I recommend a miniature type) and a little hookup wire. First, unscrew the screws on the bottom of the fone and remove the cover. You will see the Touch Tone pad on a supporting bracket. Loosen the two screws on either side of the pad and gently lift it from the supporting bracket being careful not to break any wires. The pad is enclosed in a clear, thin plastic snap case. Remove the top section of the case and put it aside. Remove the bottom section and allow it to rest on the wires leading to the Touch Tone pad.

The pad (bottom up) should look like the accompanying diagram. Locate the two toroid transformers that produce the Touch Tone frequencies. They are large and donut shaped (labelled 2597 AS in diagram). Remove the protector on the left one, if present. On the bottom edge of the pad about one inch down from the toroid transformers, you will see a three pole terminal strip (labelled 1 in diagram). On some Touch Tone fones, the terminal strips are replaced by a yellow-orange wire. If you have one of these, just cut the wire in the middle, strip both ends, and these will correspond to the normally used terminals. Locate the terminal on the left and separate the two pieces of ??? metal, one on top on one on the bottom, being very careful not to break them off. Cut two one foot pieces of insulated wire and strip a little from both ends on each wire. Now take one and solder it to the bottom half of the separated terminals. Now take the other piece and solder it to the top half of the terminal, the top half being that side with the toroid transformers. Label the two wires "A" and "B" with pieces of tape with the top one being "A".

There should be two strips of five solder terminals, one on top and one on the bottom of the left toroid transformer. We are interested in the bottom one. The fourth terminal from the left (labelled 2 in diagram) is the tap for 1633 (Pa Bell uses "standardized" components for all Touch Tone encoders whether 4x4 or the normal 3x4, so the 1633 tap is there but not used). Cut another piece of wire (about one foot) and strip both ends. Solder one end of this wire to the 4th terminal, being careful that you get a good solder joint and that you are not touching any of the other terminals. Label this wire "C" and run wires "A", "B", and "C" through the bottom plastic case along with all the other wires. Replace the plastic protector on the toroid transformer (if there was one) and snap the plastic pad enclosing case back around the Touch Tone pad. Take the pad and put it back in position on its support bracket and tighten the two holding screws.

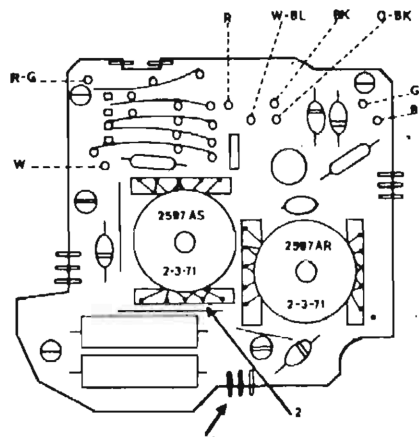
Install the SPDT switch anywhere on the telephone cover where it will not get in the way of components inside the phone. A good place has been found to be the little compartment under the hook switch on the case, looking from the back. Now take your wire labelled "A" and solder it to the middle terminal of the SPDT switch. Now take the other wires "B" and "C" and solder them to the left and right side of the switch. Put the cover back on the fone and hook up the phone to your line. If all went well, you should have the normal 3x4 touch tones with the switch in one position and when flipped to the other position, your last column of 4 buttons becomes the 1633 tones! If it doesn't work, check your solder connections to the switch and make sure you have the correct wire connected to the middle terminal of the SPDT switch. The connection to the switch is the most commonly made mistake.

When placing a regular call, remember to use the regular 1477 tone instead of 1633 or you will get a reorder. If this happens enough, Pa Bell might send a repair man to see what is wrong with your line. If you have any questions, comments, etc., please feel free to write me, The Magician, c/o TAP.



SUMMER 1980 NO. 63

1209	1336	1477	1633
697	1	2	3
770	4	5	6
852	7	8	9
941	*	0	*
			F0
			F
			I
			P



PURDUE PHONE PAD IN LA

by Cheshire

A meeting of Phone Phreaks, referring themselves as "Communications hobbyists" met on a quiet afternoon in January in the Los Angeles area to discuss the future of access to the telephone network by the hobbyists (phreaks). Speakers came from as far as the Midwest, and the East Coast to attend the gathering. The meeting was by invitation only, so many phreaks who live on the west coast may not even have found out that it was held.

Captain Centrex, a 41 year old phone phreak, hosted the conference. He discussed new Bell switching computers which are putting the phreaks out of the "free phone" area of calling around the world for a dime (and getting the dime back). Cheshire Catalyst, of the New York City based TAP newsletter talked about access to the world telex network, and about equipment displays at a telephone industry trade conference in Geneva, Switzerland he attended last September. The gist of the gathering was that computers are taking over the telephone switching world, and the phreaks had better "get digital" if they still want to rule the (micro)waves.

A few Bell System films were shown. The best of these was one called "The Bell Machine" which is highly recommended for those people interested in switching equipment. It doesn't get technical, but it is a pretty film. "Priority Immediate", a film about the fire in a New York Telephone switching center was also highly rated by the critical viewers in the audience. These films are available for civic groups, and electronic hobbyist clubs. Just call your local telephone company business office for information on how to borrow these films. Most independent telecops can get these, or other films, for you as well.

CONSUMERTRONICS CO.

LETTERS FROM READERS

1 M. E. L. 1000
1 A. 1000

2011 CRESCENT DRIVE • P.O. BOX 475
ALAMOGORDO NEW MEXICO 88310

Dear Tom,

I received your last issue of TAP. Thank you very much for your placement of REBEL in the ad flyers. At least 12 new customers to date directly due to that ad. Bringing our total REBEL subscriptions to just under 200 (and growing fast).

I have received copies of our last two issues. You may reprint any part of them but add: "From W.H.H., P.O. Box 475, Alamogordo, NM 88310." From time to time, I will insert a little blurb on TAP (free of charge) because I do want to see you survive.

I am very sorry to hear that your subscriptions are dying. I think that a big problem is that your stuff is too esoteric. I am a MSBE with plenty of applications experience and I still get low on some of your articles. And, unfortunately, most people don't share your strong dislike for Ma Bell. The 1960s are gone - this is Looking-For-No-1 time! I have found that total obsession with the utilities has limited future. My main bag now is to sell SURVIVAL! I still get in my kicks against the rip-off utilities to get my rocks off but the main objective is easy and inexpensive to apply survival information. I have found that many Americans are expecting A Armed Rebellion, B Total Economic Collapse, and/or C Foreign Power Invasion, very soon. So don't be alarmed if they don't get excited about credit card calls and bond boxes.

Change your format to reflect this change in people's interests and objectives, and you will be stronger than ever, as unusual and hard-to-find info. is now very much in demand. Also, consider selling books and pamphlets of your own creations, just as I do. They do provide financial security different than that provided by a periodical.

I understand that you are considering doing your own printing. All my printing is done commercially. It is expensive. Six months ago, I invested heavily into printing equipment and supplies to do it myself - an expensive disaster. The press turned out to be a piece of junk (\$1,300?), which I've mostly repaired myself. AB DICK parts and labor are astronomical (when you can find them!). Supplies are numerous and expensive. It's a real art to work, maintain and repair offset presses, not to mention very time consuming and just plain hard work. Rubber galleys are fine until you lose a piece of paper in the form roller, which you have to dig out with your fingernails - inky hands may ruin your meals and sex life. And a press buys you very little without an offset platemaker, paper jigger, folding machine, power stapler, etc., etc. - it gets out of hand FAST!

I have found that doing my own layout work very interesting and profitable. I have a Verity 820 Handwriter and 70 fonts, a 96C Olivetti proportional typesetter (this letter and a light table of my own construction). I can make you some headlines free of charge, just let me know and I'll send you a copy of the fonts that I have. I understand that Zenith International, 41 E. 20th St., NYC has Veritypress at reasonable price (conditions unknown).

Well, take care. Have a good day. Keep on trucking.

Best Regards,

John J. Williams

John J. Williams, President

Visit Phon City

by Chesnise



Many people visit the Big Apple each year. Many attend the various trade shows and conferences. Others, come to catch a Broadway show or visit the other wonderful sights the City has to offer. A few come in on Friday evenings from 5 to 8 PM to snare the companionship of fellow technology freaks who enjoy good food, and good beer in the back of a small restaurant on West 4th St. (between 6th and 7th avenues) in world famous Greenwich Village.

Dionysia is a small Greek restaurant with a host of middle east specialties. (For those into Greek, but not Greek food, Christopher at 15 is a block north, but don't expect us to join you.) A group of TAP's writers and editors gather here in the back room, and give information on the latest in satellite technology and telephone switching, most of which never sees print in TAP for various reasons. (lots of it is pure bull shit)

If you've wanted to come in to visit, out never had the chance. My wife and I only will try to determine for you to stay with one of TAP's motley crew of irregulars. Don't expect much more than a piece of floor and a sleeping bag, but you're certainly welcome to what beer is available. Write to TAP, care of the address, and give the envelope. (Including 1000 for the company is sending you what you just want to not together and end, or have a local show you around, just set in touch. A few of us have been visiting Europe and have been on expense account funds to prove survival and to the delight of all.

Hope to see you soon.

Talk back -even to Phone Co.

LETTERS ARE RECEIVED from practically every place with regular companies about local phone companies. It's a typical case of information in their service in the past few years which they have been told that they are a "real" audience that can't just walk down the street to a competitor phone company.

What can you do about it? But it's not a question that a statement of organizational structure to whether rate and service policies the company chooses to establish.

Like most consumers, I believe our phone system is better than any other in the world. Like an increasing number of consumers, I think it could be better than it is. Any utility, especially one which consumes an area, must be fully accountable to its customers. If a utility commission does not initiate the questioning of whether any rate schedule or company practice is in the consumer interest, then the consumers themselves must initiate such questioning and make the regulatory commission move.

That's happened in many communities where independent consumer groups which have made phone company their top issue (sometimes only) priority have organized their complaints and won increased consumer protection. For those who are interested from the account story of a consumer group in Massachusetts.

A three-year campaign initiated by a group called "Rate Share" and signed by the Public Law Institute, the Public Interest Group, the Legality Center of Older Americans, and the attorney general's office, the Massachusetts Department of Public Utilities, has resulted in the state's first law which provides increased protection against some of the most common complaints. Here's what the Massachusetts Board of Public Utilities has to say for them:

No deposits are required for most customers, although many are required for those who are not. Only those who are not required to pay deposits are required to pay a deposit. Customers who have already paid a deposit must have it returned with interest.

Customers with complaints about any billing issue are entitled to the Department of Public Utilities, which will serve as impartial arbiter.

Customers will have 30 days to pay a bill and if still not paid on the 30th day, the company must notify customer 10 days in advance if a shutoff of service is planned.

Customers falling behind in bill payments may arrange for installment payments over a maximum eight-month period with interest charges.

Customers having difficulty paying bills because of serious illness, verified by a doctor, or for other personal circumstances for which phone service is vital, have the right to a short or postponement of 30 days, renewable if necessary for an additional 30 days.

Businesses can now be shut off in a home where everyone is 65 or older. The company must notify the Department of Elder Affairs, and receive written permission or proceed.

The company must inform new customers, or old customers seeking new service, about all the possible options, including the least expensive option.

Information on all customer rights must appear in the first of the phone book.

The state themselves are important. But even more important are the information which such information can give to the public. It is the company and the utility commission. And that's how. You might call it the "Bureau of Telephone" Party. And you know what that started.

Gronny's pot luck: 60 days in jail

Venture Capital (1979) - Low Profile: Mr. Gronny, a 36C Olivetti proportional typesetter (this letter and a light table of my own construction). I can make you some headlines free of charge, just let me know and I'll send you a copy of the fonts that I have. I understand that Zenith International, 41 E. 20th St., NYC has Veritypress at reasonable price (conditions unknown).

Best Regards,

Ma Bell, an owner of a small business is really burned by Ma Bell. He claims they double or triple the number of times he calls frequently called numbers. At 9c per call, they manage to get a pretty good bill worked up.

My boss once carried his complaints up into higher management levels and once was promised by someone high up that he would have an automatic call number installed. This never materialized and the lower levels claim no such device exists.

I would like to ask TAP readers what they know about this subject. If they don't exist, how hard would it be to develop them? There might be a pretty good market for such items.

I am an electrical and often install conduit for telephone and other communication systems. These Bell employees are the most atrocious bunch of loafers you're ever going to see. Not only that, they've been known to sabotage the equipment of independent wiring companies.

I have personal knowledge that Bell will ignore certain rules when they to some types of installation but make the independents follow every rule to the letter.

The Omega Man

RAO Credit Card Codes

201 093,094,101	408 293	618 204,547
202 032	412 030,042	701 133
203 020	413 009	702 176
204 456	414 088,089	703 033
205 054	415 158,159,167	704 319
206 163	416 476,478	705 477
207 003	417 157	707 223
208 039,120	418 505	712 311
209 254,289	419 057,248	713 105,151
212 017,021,023, 072,074	501 147	714 164,182
213 066,183,184	502 048	715 202,330
214 149	503 131	716 026
215 041,043	504 046	717 027,028
216 050,082	505 102	801 155
217 081,314	506 451	802 002
218 236	507 310	803 189
219 320	509 128	804 257
301 011	512 146,152	805 252,255
302 010	513 077,185	806 103
303 153	514 470,472	807 481
304 034	515 134	808 461
305 044,180	516 127	809 490 (P.R.)
306 457	517 095	812 321
307 137	518 024	813 531
308 237	519 484	814 208,307
309 203,260,277	601 059	815 087
312 086,097,098, 196,234	602 064	816 144
313 083,096	603 004	817 150
314 143,177,251	604 493	819 474
315 303	605 138	901 187
316 140	606 317	902 452
317 080	607 025	904 056
318 051	608 201,329	906 295
319 312	609 101	907 492
401 019	612 126	912 316
402 139	613 473	913 145
403 454	614 079	914 069
404 035,063	615 047	915 221
405 148,215	616 084	916 160
406 154	617 001,006,007, 008	918 141 919 036

A Telephone Credit Card Number is 14 digits, divided 3-2-4-4-5. The first ten digits are the phone number being charged to. The next three are a billing code (the Regional Accounting Office, or RAO). A list of the major ones is at the left. The final digit is for verification, and changes annually. For 1980 (1/1/80 to 1/31/81), the verification digit has to match the SIXTH digit of the charged-to number. The code is:

6th digit: 1 2 3 4 5 6 7 8 9 0
check digit: 4 8 1 5 9 2 6 0 3 7

Example: phone number 202-456-1414
Credit Card Number 202-456-1414-0324



(To use the Credit Card Number from outside North America, an International Prefix must be added. The 1980 International Prefix is IX.)

No set of rules can take the place of common sense, but there are a few basics to using Telephone Credit Card Numbers.

Make Credit Card calls ONLY from pay phones (there's no record of the originating phone), and at random times and locations. Always be courteous to the operator (think how much company shit s/he has to put up with), and remember that s/he may listen to some of your conversation.

It is harder to trace someone who never uses the same Credit Card Number twice. Long Distance Information is free, and with a little imagination, there are endless possibilities. Since the call does end up on someone's phone bill, don't make up numbers at random.

When a fraudulent Credit Card call is reported to company Security, they may hassle the recipient of the call. If the call went through an institutional switchboard, they're stuck. If the call went to an individual, they may call and try lies, sympathy, intimidation, or whatever may get money or information. Or they may just stick the call onto the bill, months later. So only call individuals who will be dumb on the phone and sharp reading their phone bills.

And not too often.

Justice Is Blind To Pickpocket

Seoul (AP)—During the trial of a woman accused of picking pockets in Taegu, South Korea, someone picked the pockets of three people in the gallery and escaped with \$65.

TRUTH IN ADVERTISING

The Tulare County, California, Sheriff's Department is looking for the man who left two officers holding the bag.

The bag, it seems, was brown paper and held horse manure... \$200 worth. Earlier, undercover agents had arranged for the purchase of two pounds of marijuana and the bag of manure was what they got.

The police may have difficulty in getting a conviction since the culprit was heard on a tape recording of the transaction clearly state, "Hey... it's good shit."

STORRS, Conn. (AP)—The University of Connecticut recently installed a \$14,000 computer system to help keep people out of the Institute of Material Sciences.

The idea was to cut down on thefts. It didn't work.

Police reported Thursday that someone stole the system from the institute's building.



"Gentlemen, I just wanted to remind you that the home office will be monitoring this on closed-circuit TV, the sales department is recording this for future use, and one of us here is an FBI agent."

Reuse of stamps costs Postal Service millions

WASHINGTON (AP)—A Nebraska senator's office aides aren't the only ones trying to save a few bucks by tearing uncanceled postage stamps off the boss' incoming mail and re-using them.

The cash-starved Postal Service says it's losing \$25 million to \$75 million a

year because of the widespread practice, which happens to be a federal crime.

Sen. Edward Zorinsky's office acknowledges doing it as a way to cut down on mailing expenses.

Carolyn Andrade, administrative assistant to the Nebraska Democrat, said

yesterday "On a slow day I get \$5 to \$10 worth of stamps off incoming letters. That's like picking up money in the street."

She contended that since the stamps apparently miss the Postal Service's cancellation machinery, they can legally be reused. She said the Postal Ser-

vices "ought to be a little more careful at looking at the thousands of dollars (in stamps) that are being used over and over again."

Postal officials say it would be impractical to try to find and prosecute the many people who re-use uncanceled stamps.

Back Issues are 50¢ each. Issue #50 is \$1.00.

Subscriptions - 10 issues - US Bulk rate \$5.

US First Class in plain sealed envelope \$7.

Canada & Mexico First Class \$7.

Foreign \$8.

IMPORTANT! Include mailing label or Xerox copy when writing to TAP about your subscription.

Electronics Courses - 50¢ each. A - DC Basics, B - AC Basics, C - Phone Basics, D - Amplifiers.

TAP Mugs - \$4.50.

TAP T-shirts - \$4.50. Specify size and color:

Medium, Large, Extra Large. Black or red T-shirt with white TAP logo.

TAP "Cracked Bell" Patch - \$1.50.

TAP "Cracked Bell" Button - 50¢ each.

TAP Cassette Tape - \$3.50. Hear Chap Crunch, Al Bell, Joe Engressia, and Bell Security Chief John Doherty.

Send cash, check, or money order to:

TAP, Room 603, 147 W. 42 St., NY 10036

63

"I always keep a supply of stimulating hands in case I see a snake... which I also keep handy." - W.C. Fields

NEW YORK (AP)—The American Telephone & Telegraph Co., citing the need for attorneys to work on mounting federal regulation, employs more full-time lawyers than the country's two largest law firms combined, according to the National Law Journal.

The weekly newspaper reports in its current issue that with 863 lawyers on its payroll, ATT is the largest private employer of full-time lawyers.

The largest private law firms were Chicago's Baker & McKenzie with 512 lawyers and New York's Shearman & Sterling with 380, the Law Journal said.

Telephone privacy detailed in booklet

NEWARK (AP)—New Jersey Bell has published a booklet that gives customers information about their rights to telephone privacy and how the company protects the privacy of its customers.

Entitled "Our Customers' Right to Telephone Privacy," the booklet details New Jersey Bell's policies in areas dealing with the privacy of telephone conversations, telephone listings, wiretapping, annoying and harassing calls, sales calls, automatic announcements and customers toll records and credit information.

Consumer Affairs
New Jersey Bell
Room 1506, 540 Broad Street
Newark, New Jersey 07101

"A woman drove me to drink and I never even had the courtesy to thank her." - W.C. Fields



PETRODOLARS, NICKELS AND QUARTERS are choking Findlay Ohio, pay phones. When the 30 Saudi Arabian students enrolled at Findlay College get homesick, they call home, at the rate of \$0.65 for the first three minutes and \$3.20 for each additional minute. That means \$64.05 worth of change for the average 20-minute Saudi chat. The college keeps clogging the collection boxes on Findlay's 23 pay phones, knocking them out of order.